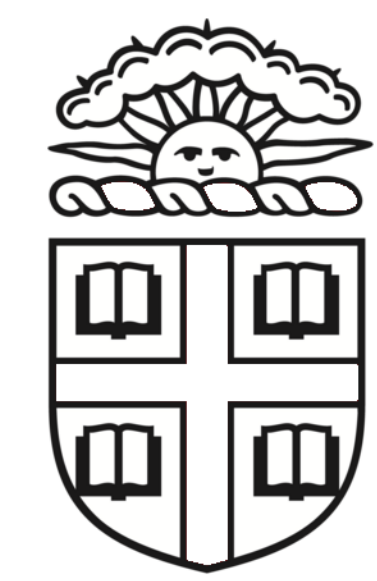


The road ahead: knowledge does not replace vision in navigational decision-making

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VSS poster 23.468

Background

Previous work has shown that navigators select routes by choosing turn options that deviate angularly less from the goal vector [1, 2]. However, as these studies involved wayfinding through sequences of occluding barriers, it is unclear if this preference would remain when route options are fully visible or fully known.

Methods

Summary

- Participants (exp. 1: n=28; exp. 2: n=29; exp. 3: n=12) navigated around courtyards in VR (Meta quest 3) within a 13mx13m space
- Starting conditions varied: turns either made in motion (on-ramp) or from standing (orientation-only)
- Total of 90 trials varied approach angle, crtyard dimensions and L/R side

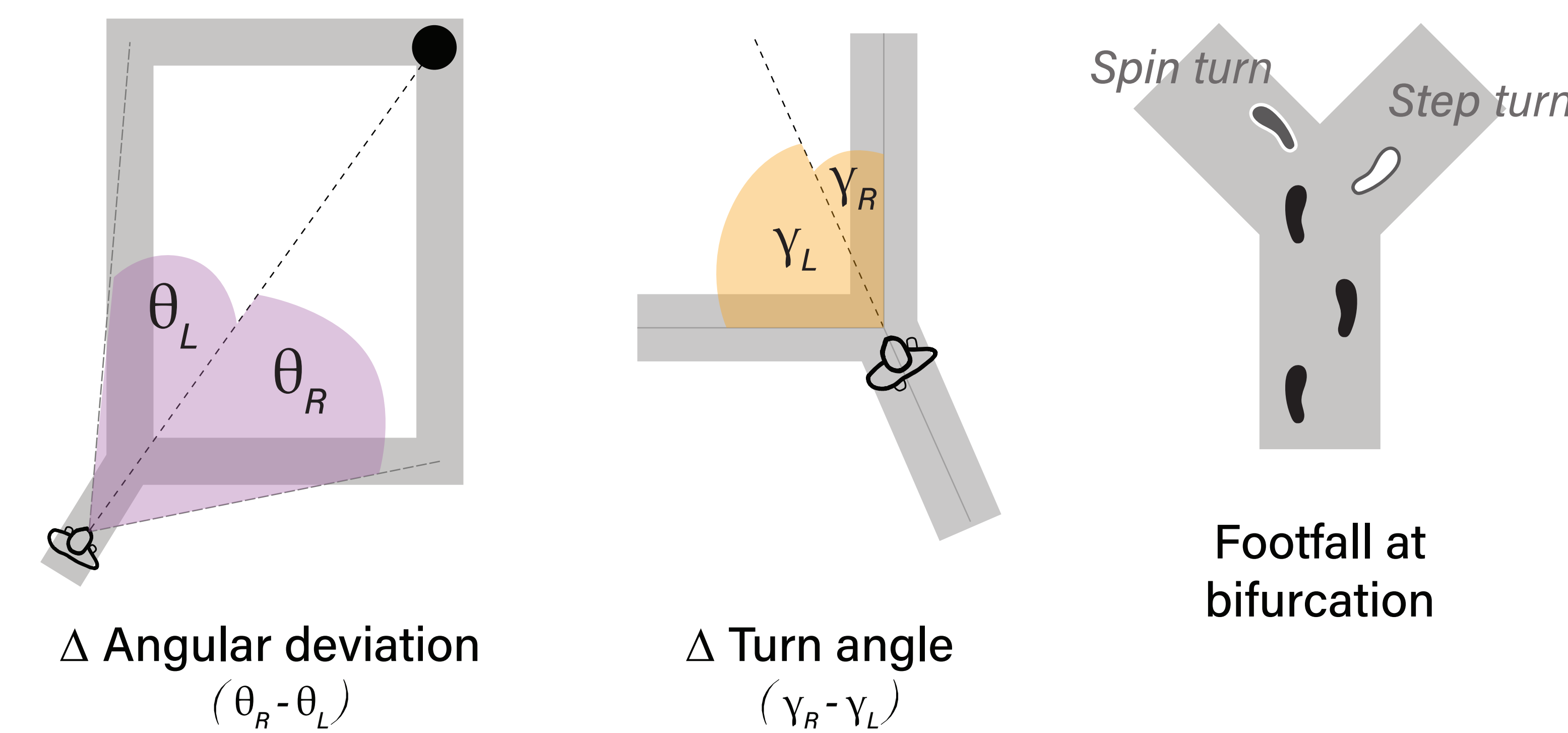
Trial structure

- Enter start pole
- Face orientation pole
- Courtyard appears
- Walk to goal pillar on sidewalk

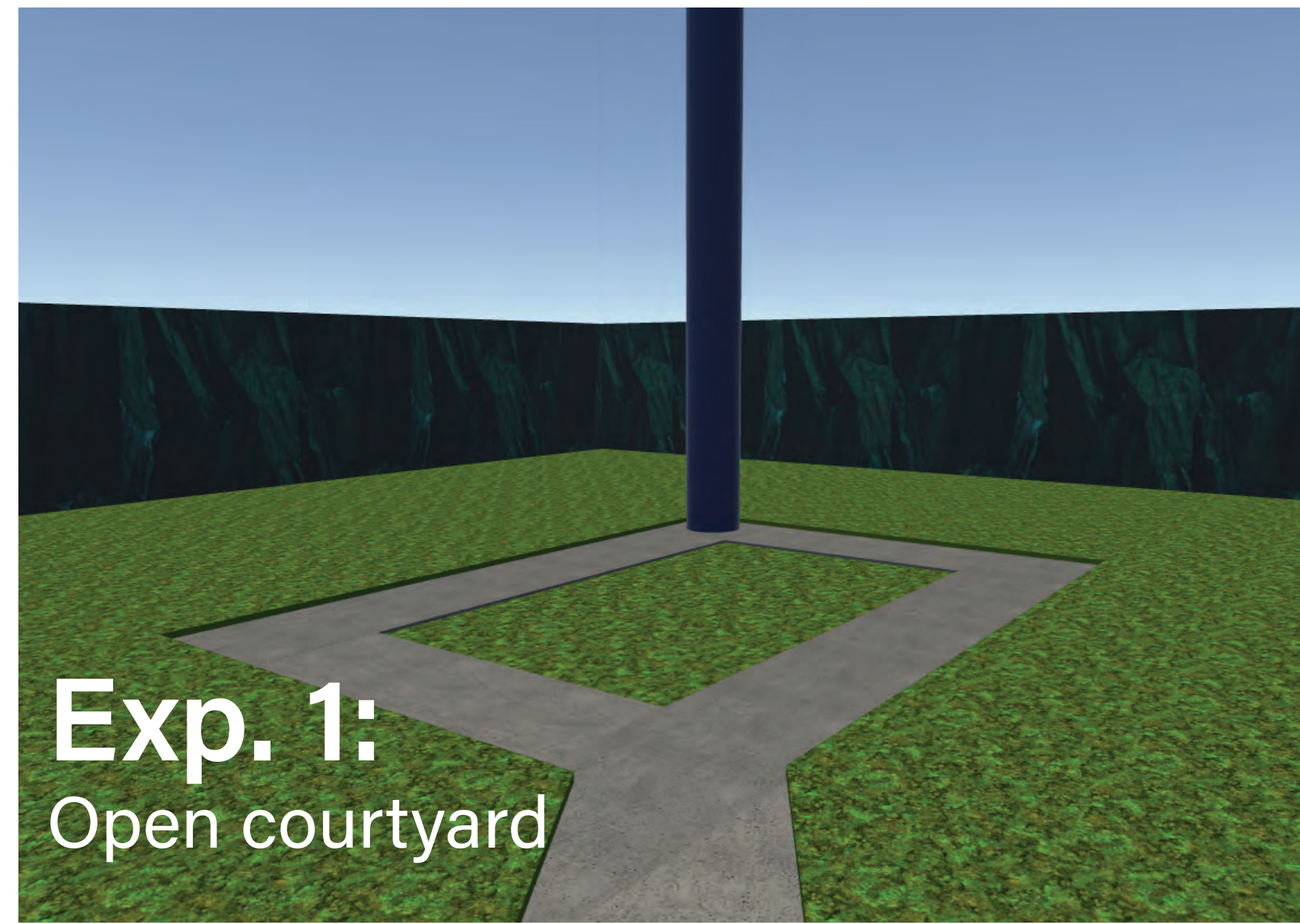
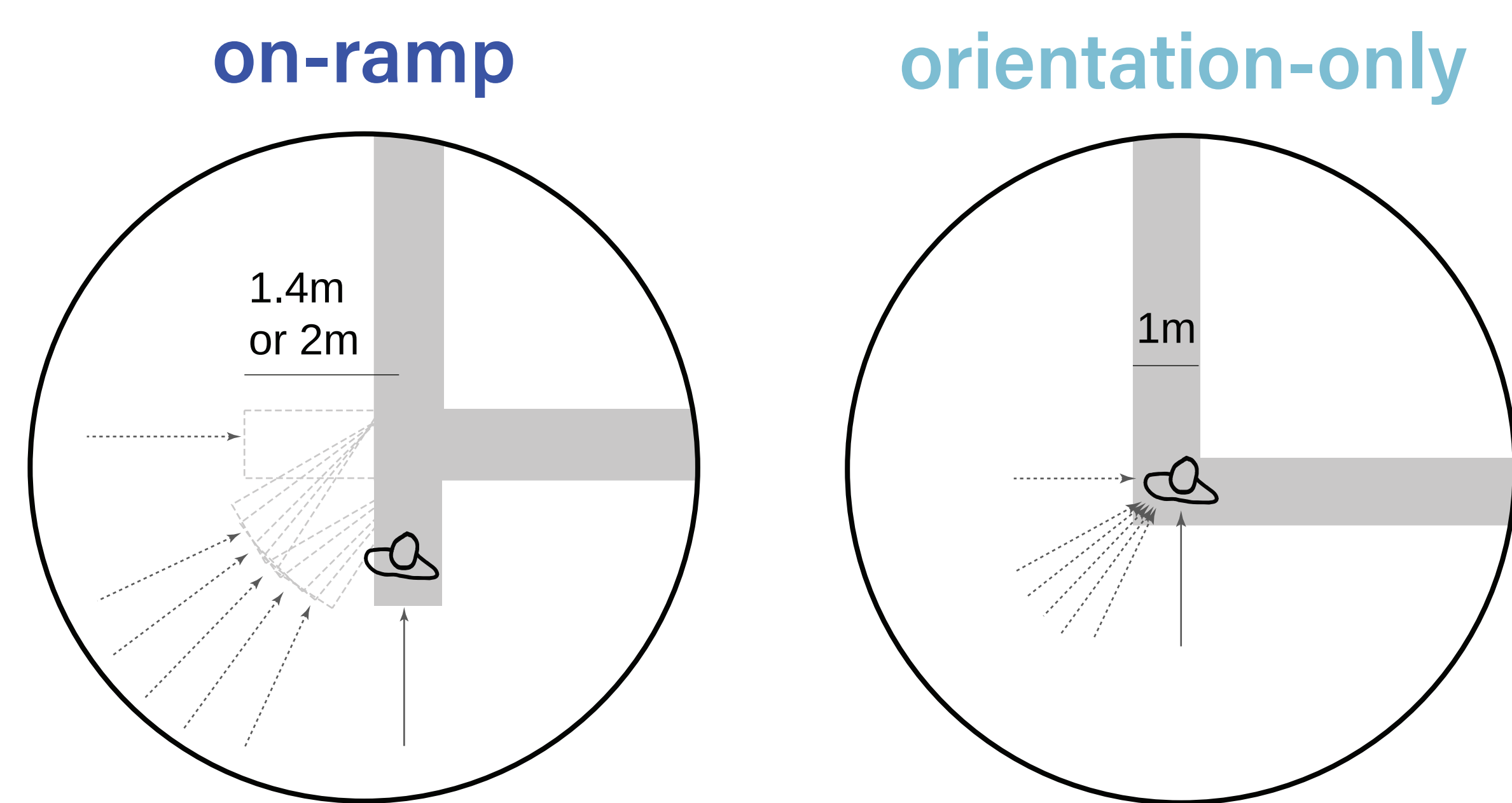
Statistical Methods

- Mixed effects logistic regression models (GLMMs) predicted turns based on fixed effects (independent vars.) + rand. intercept
- Chi-square tests evaluated whether variables were contributive

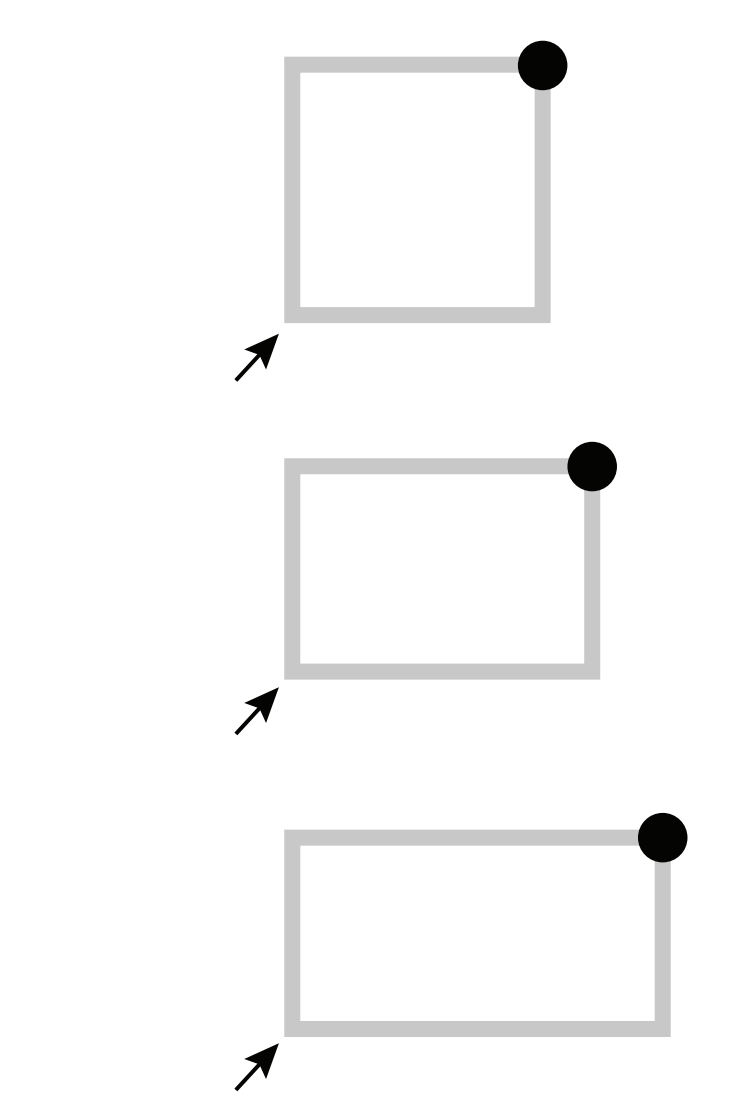
Independent variables



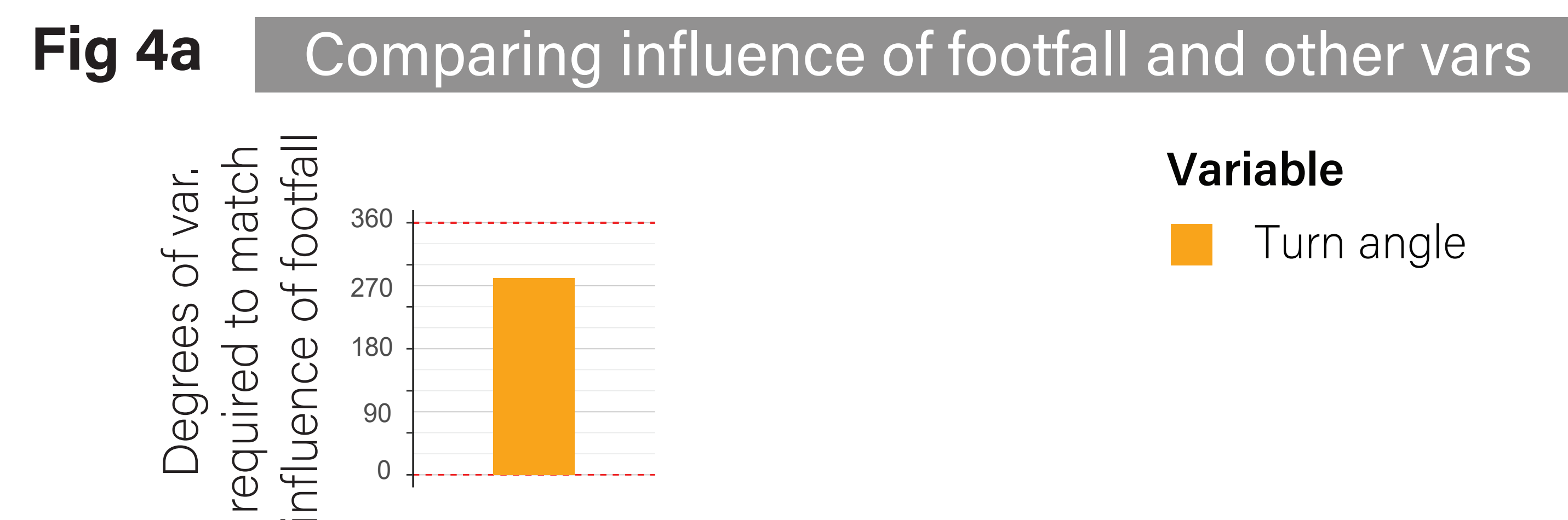
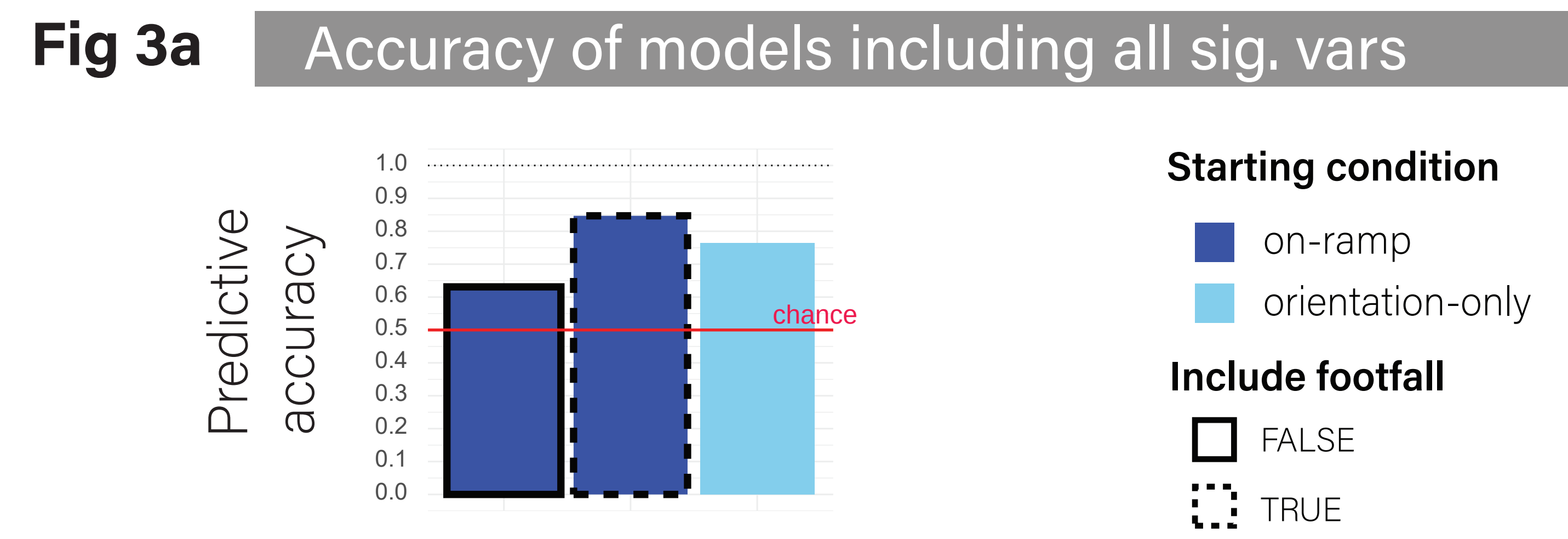
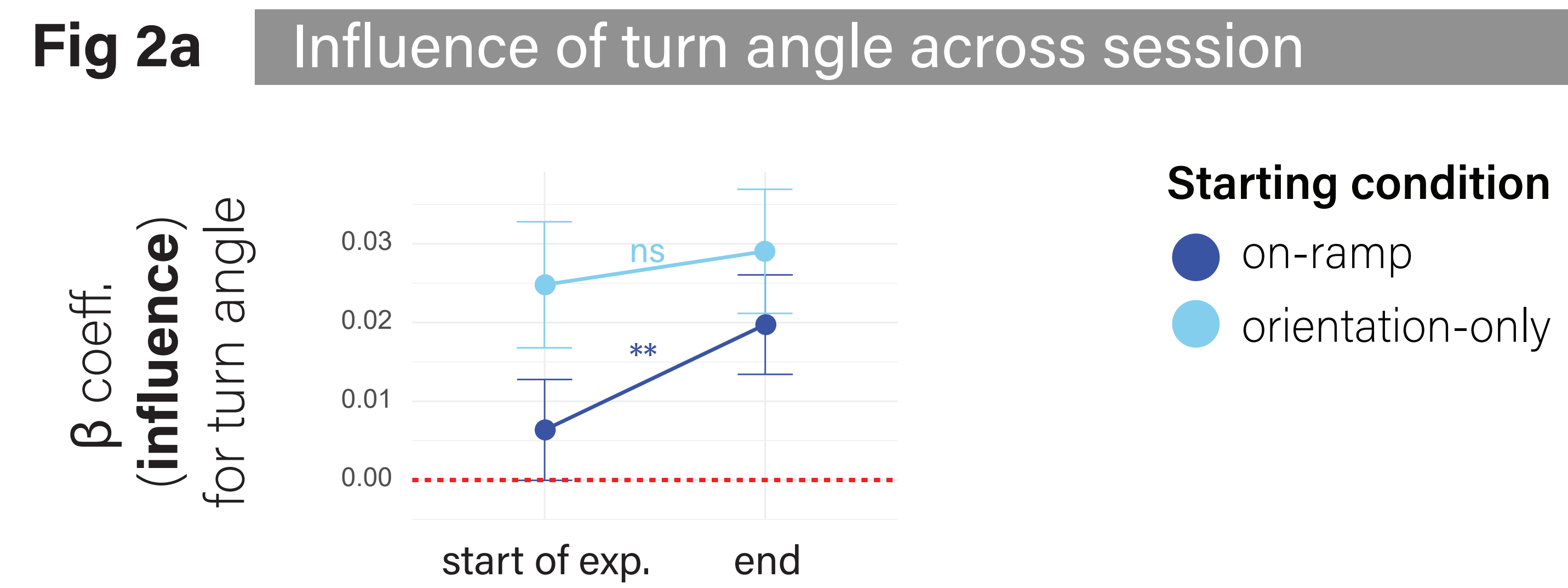
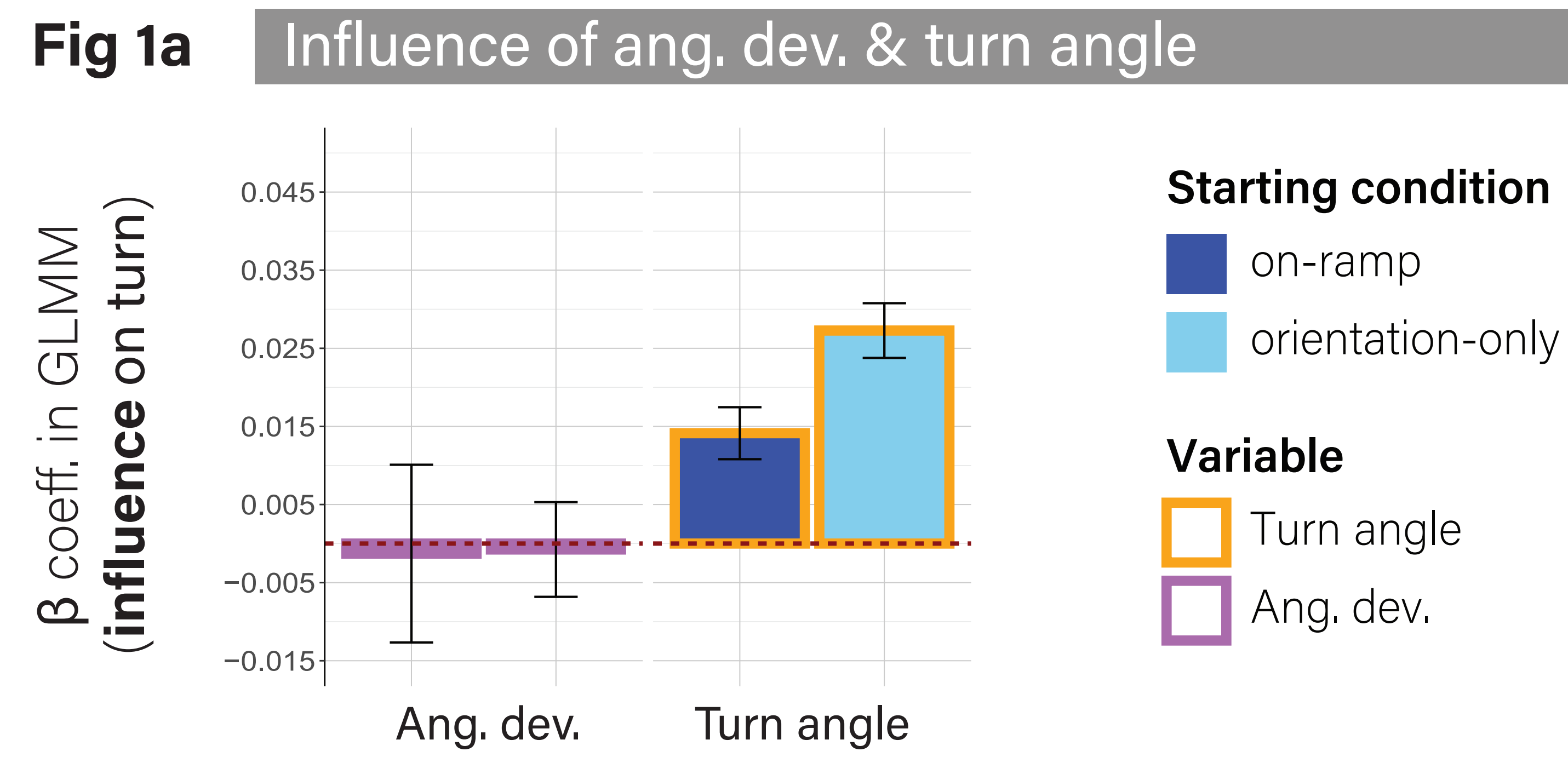
Starting conditions



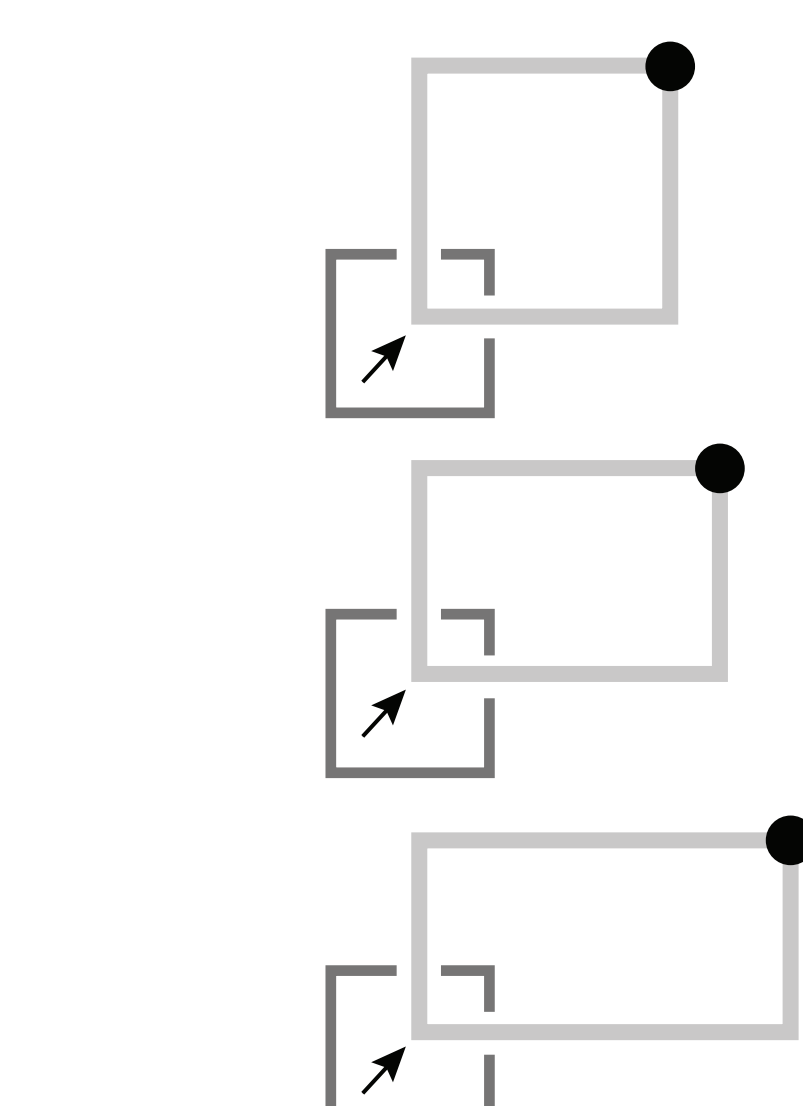
Exp. 1:
Open courtyard



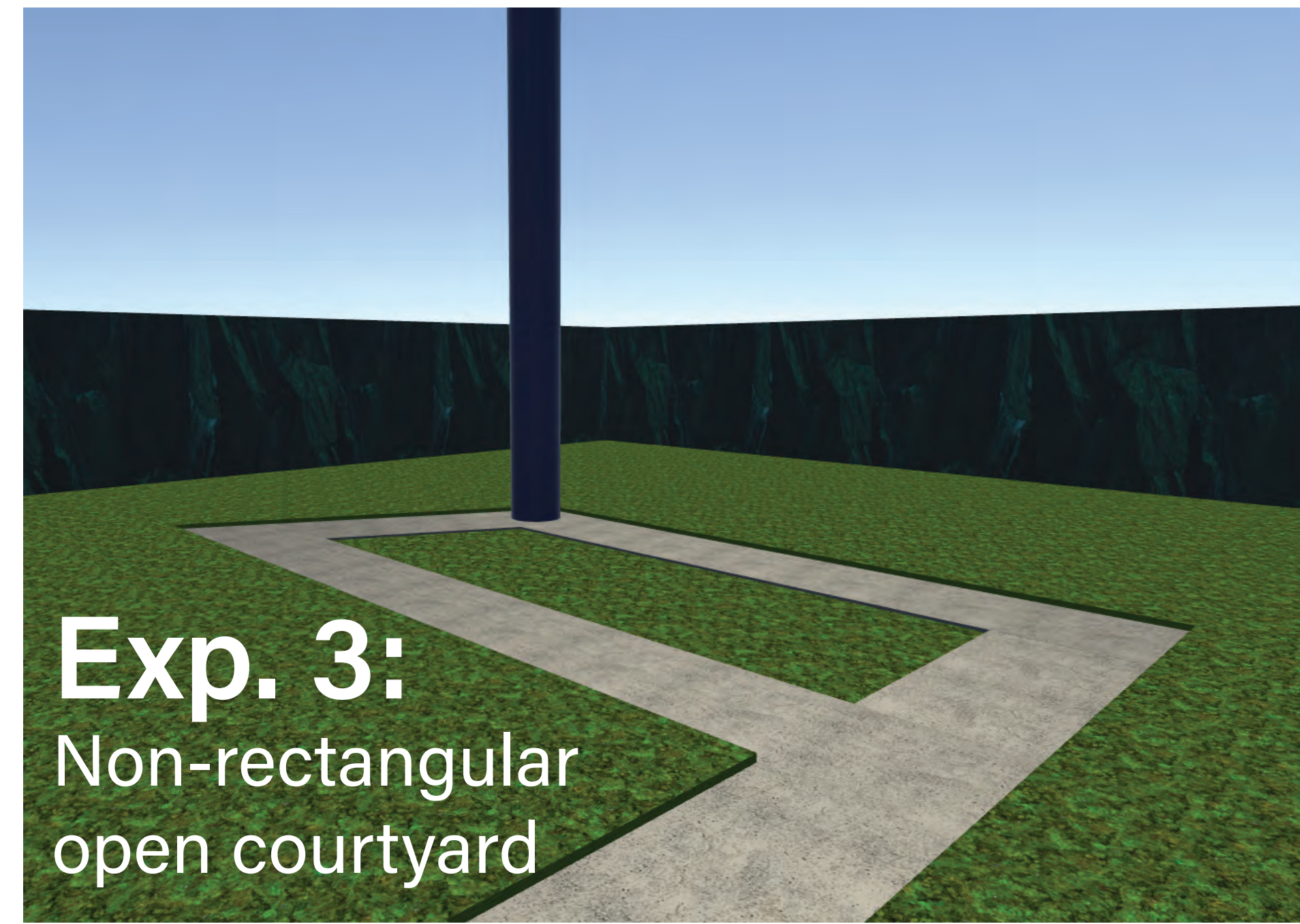
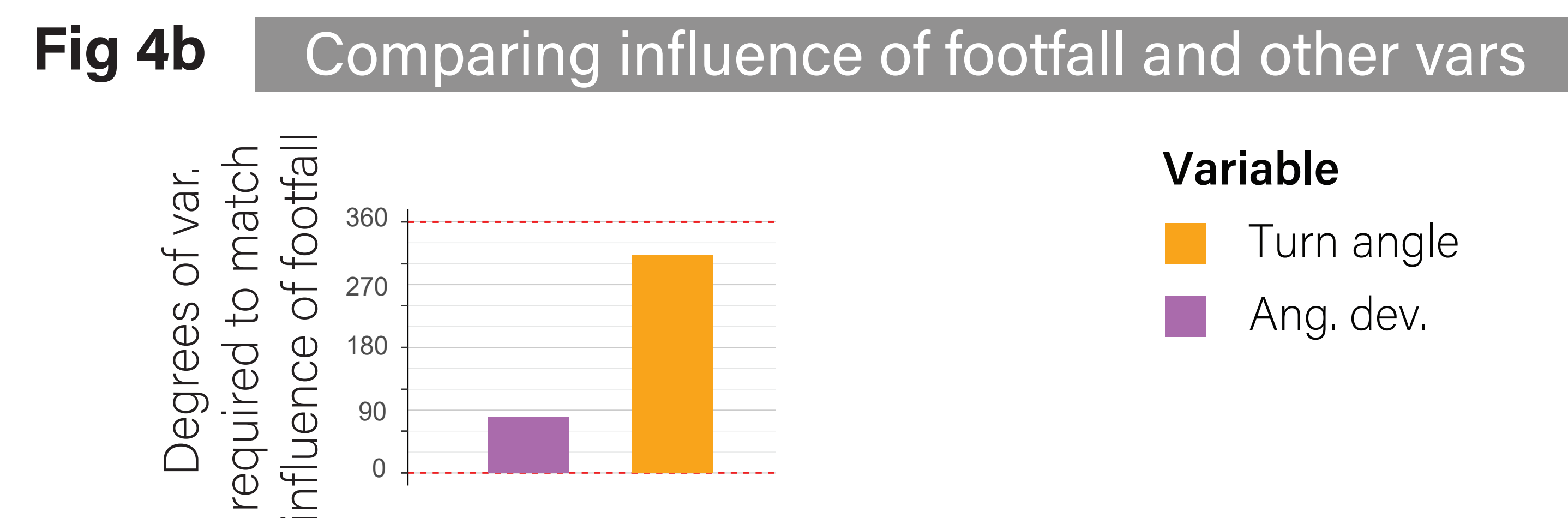
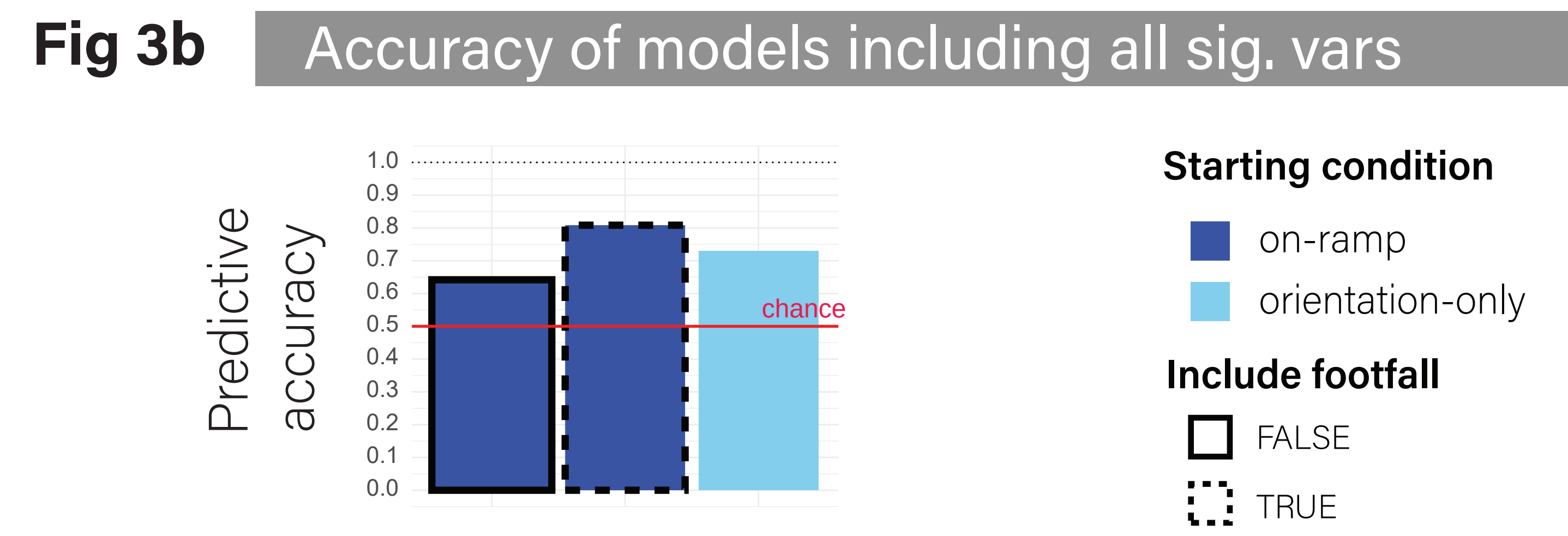
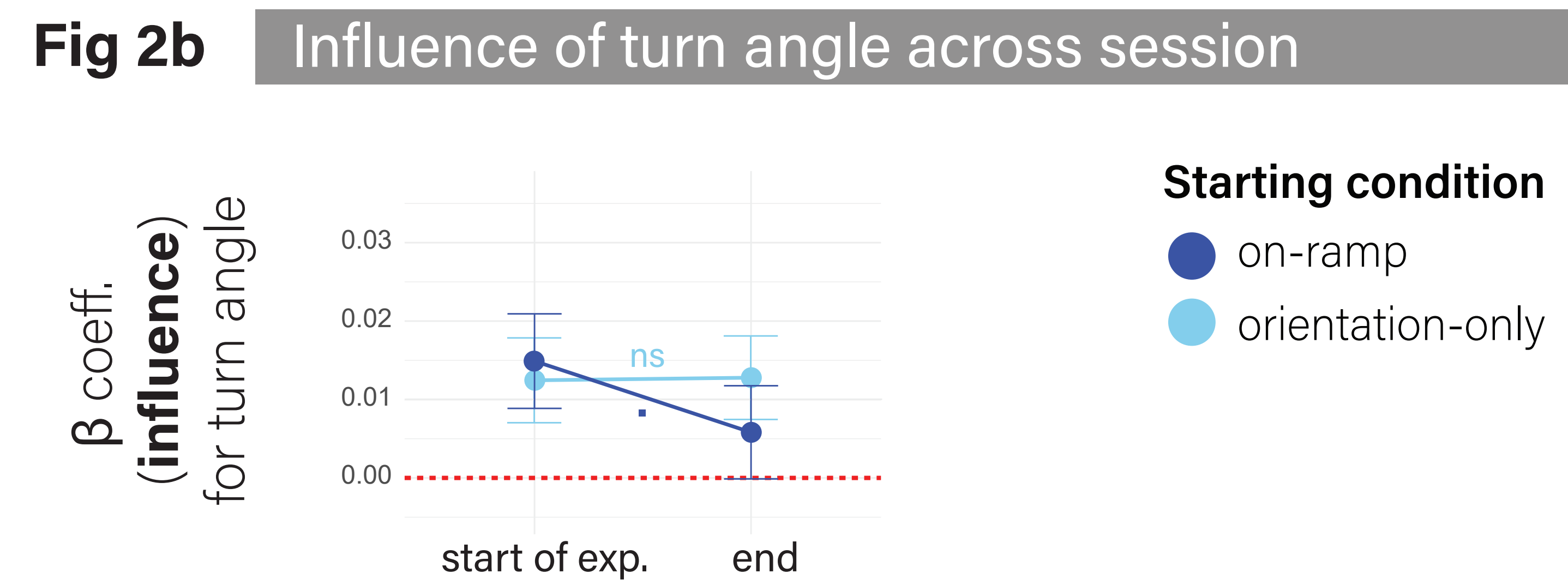
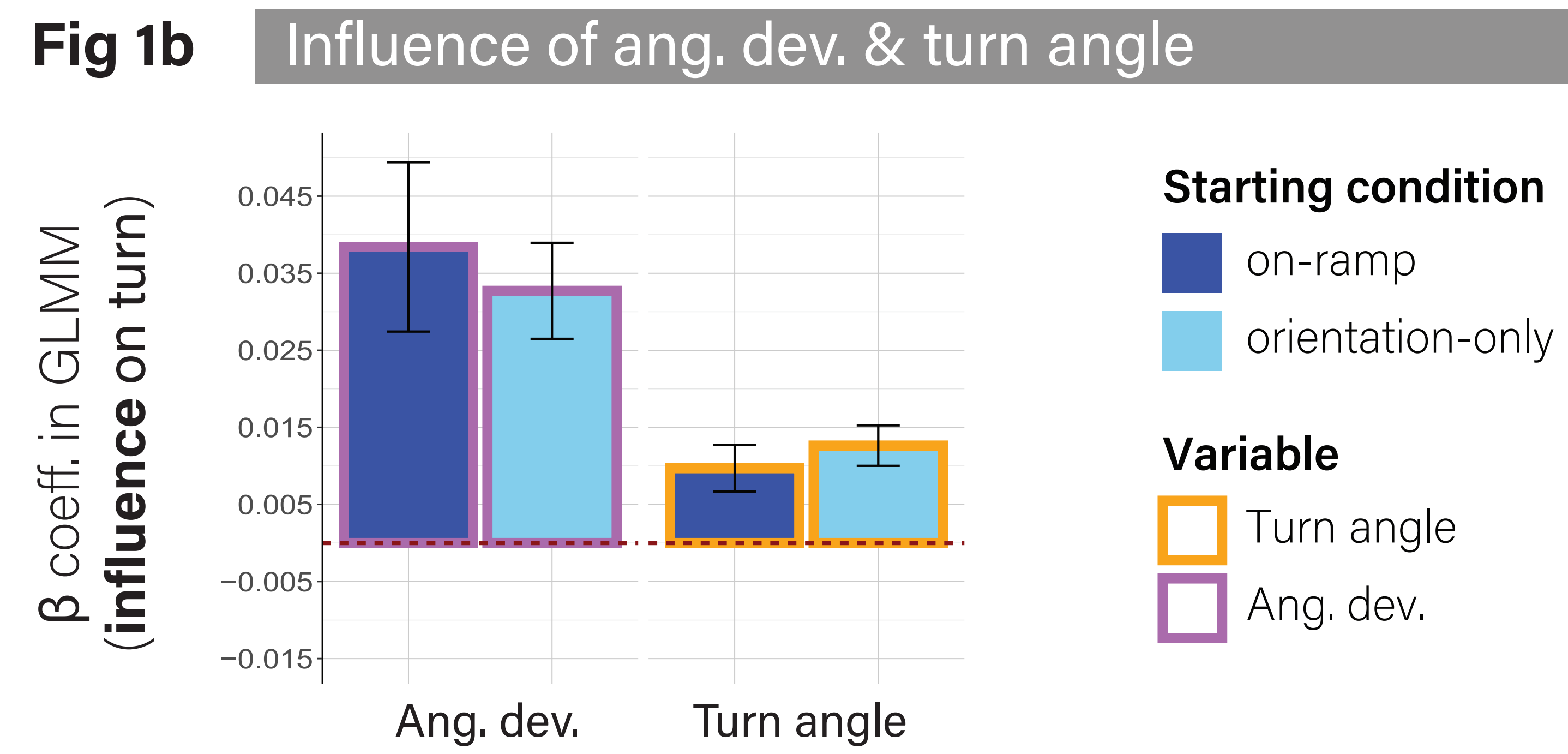
Courtyard types



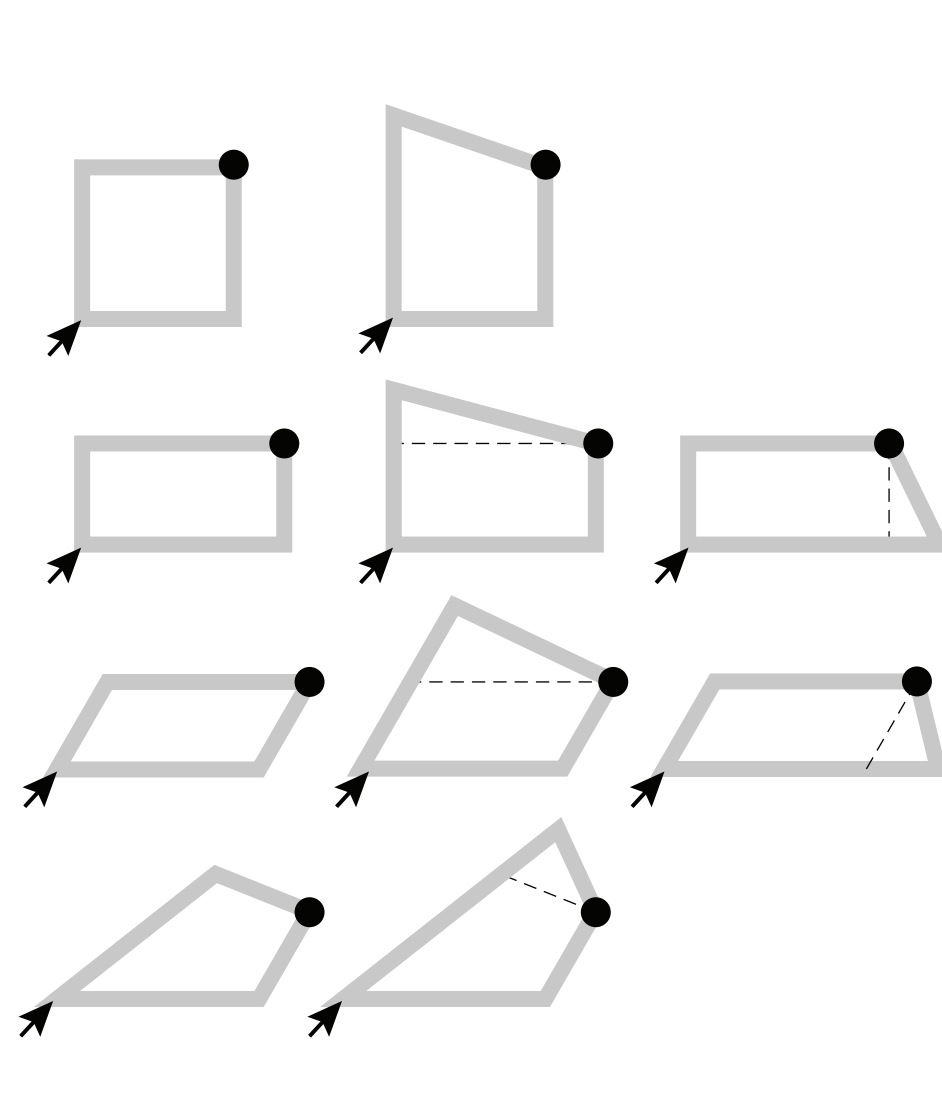
Exp. 2:
Blocked courtyard



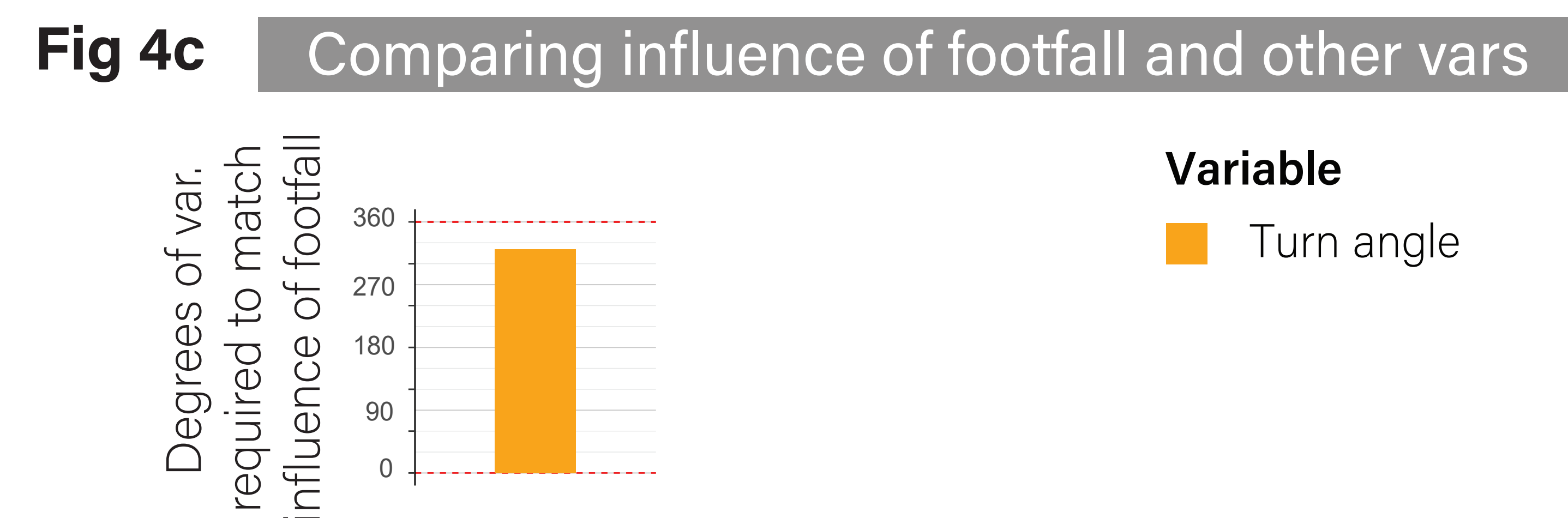
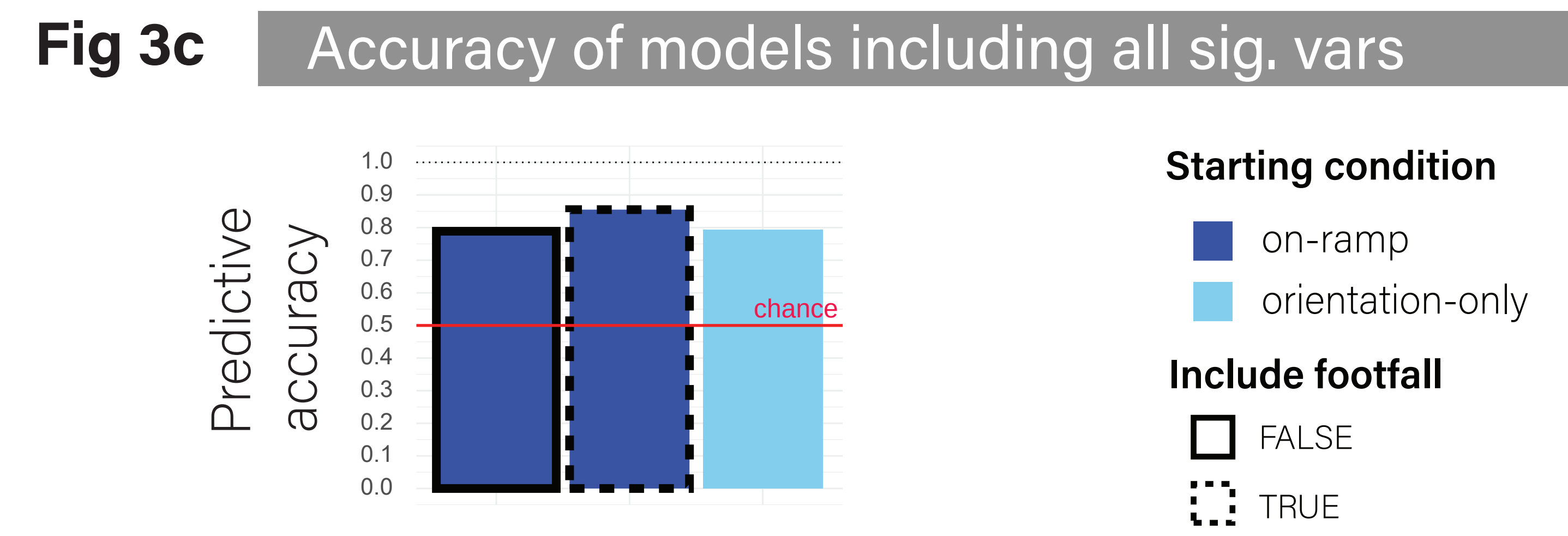
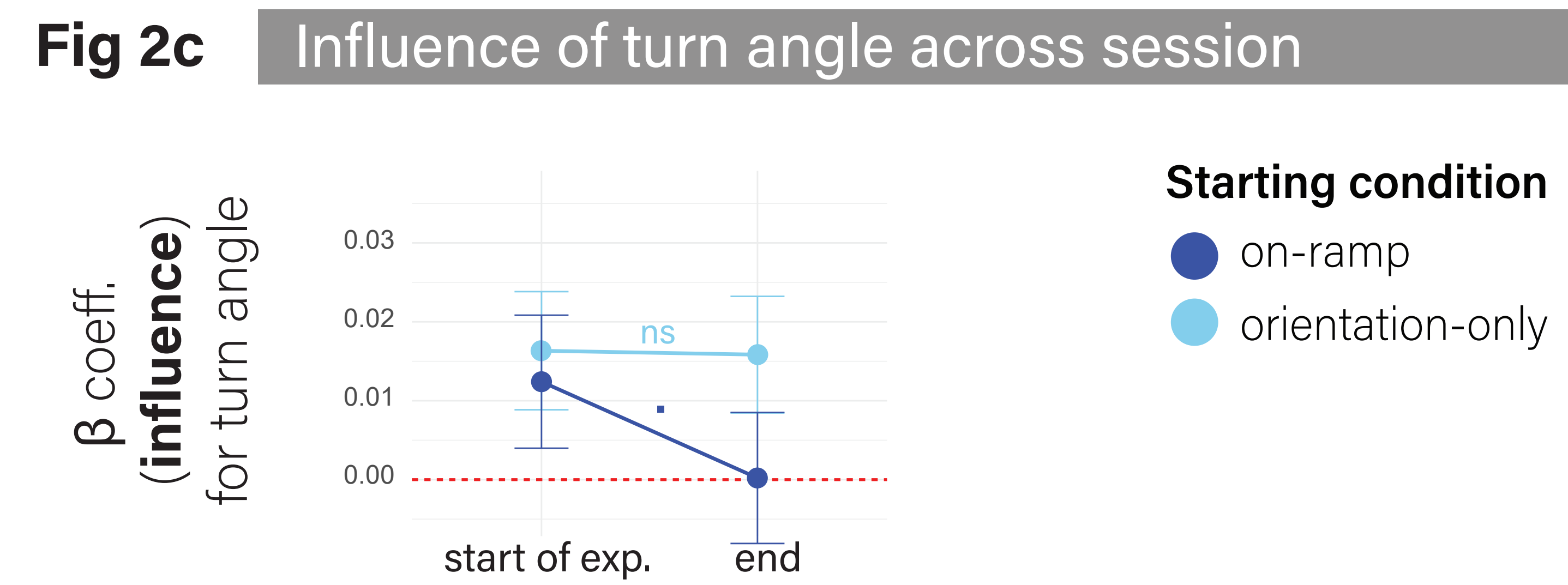
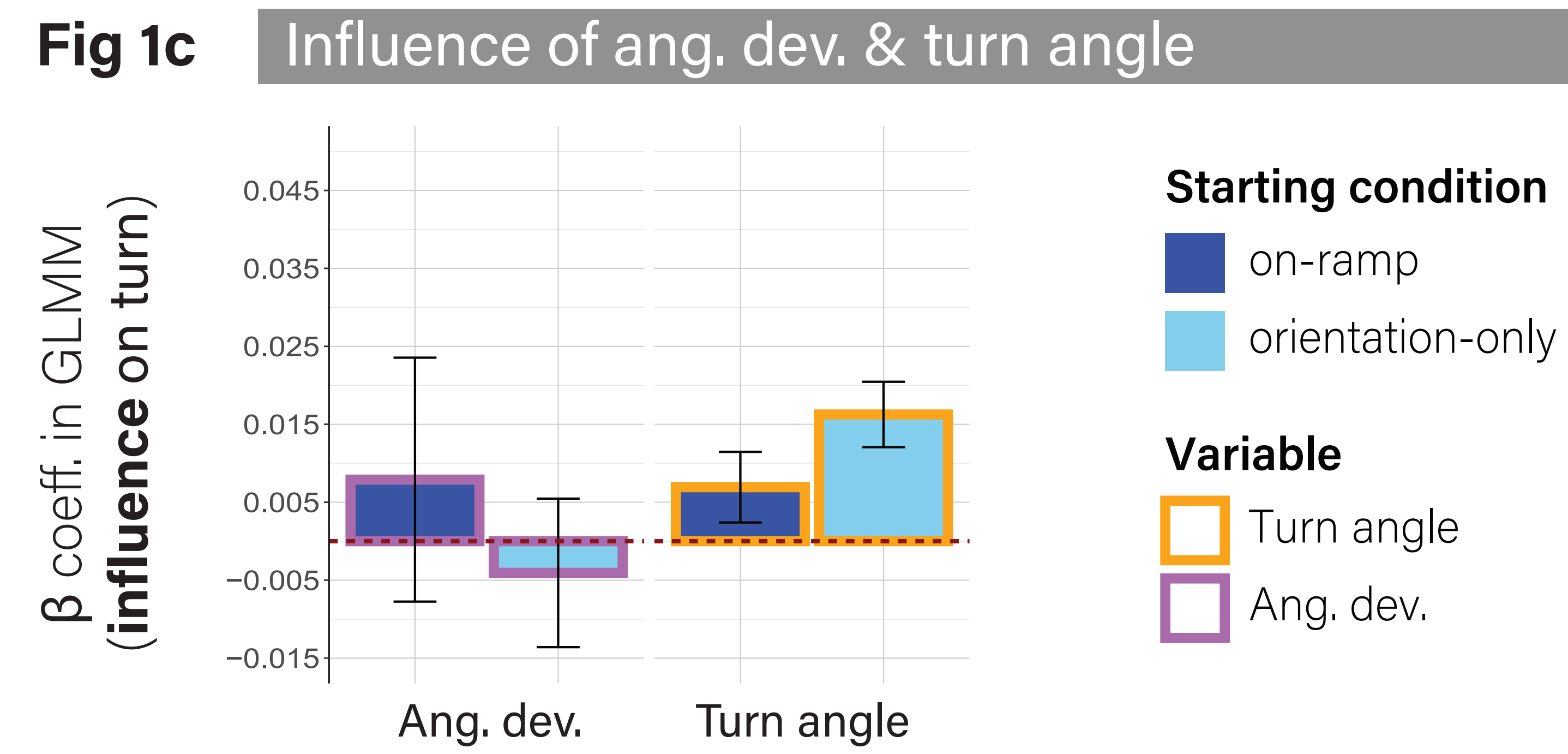
Courtyard types



Exp. 3:
Non-rectangular
open courtyard



Courtyard types



Conclusions

The preference of navigators to remain oriented with the vector to their goal is only present when the candidate routes are not visible. When choosing between routes that are visible, turn decisions are made on a basis of energetic efficiency, including reducing route length, turn angle, and choosing 'step turns' [3].

References

- [1] Engstrom, C. & Warren, W. H. (2024, May 18). *Global Route Selection using Local Visual Information* [Poster presentation]. The Annual Meeting for the Vision Sciences Society (VSS) 2024, St. Pete Beach, FL, USA.
- [2] Engstrom, C. & Warren, W. H. (2025, August 26). *A simple visual heuristic allows maze navigators to choose shorter routes under combinatorial uncertainty* [Poster presentation]. The European Conference on Visual Perception (ECVP) 2025, Mainz, Germany.
- [3] Hase, K., & Stein, R. B. (1999). Turning strategies during human walking. *Journal of neurophysiology*, 81(6), 2914–2922.